

Aryan Bhusari

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

August 2024 – May 2026

- GPA: 3.66 | Selected Coursework: Adversarial & Trustworthy Foundation Models, Applied NLP, Deep Learning, Advanced CV, ML for Data Science, Analysis of Algorithms, Information Retrieval and Web Search Engines, Database Systems

Maharashtra Institute of Technology

Pune, India

Bachelor of Technology in Computer Science and Engineering

October 2020 – May 2024

- CGPA: 9.36 on a scale of 10 | Selected Coursework: Mathematics, Big Data Analytics

EXPERIENCE

Arm Inc.

Austin, TX

Intern, Machine Learning Software Engineer

May 2025 – August 2025

- Researched and benchmarked distributed inference for large quantized LLMs (up to 405B parameters) across AWS, Azure, and GCP, contributing a new benchmark to a Terraform-based infrastructure tool and modifying llama.cpp (C/C++) source to expose RPC communication for empirical analysis
- Conducted bottleneck analysis with Linux perf, Wireshark, and tcpdump, identifying CPU core underutilization and implementing fixes for a 100% throughput improvement; characterized scaling behavior across instance types, batch sizes, and prompt lengths

Centre for Development of Advanced Computing

Pune, India

Deep Learning Intern

July 2023 – December 2023

- Developed a deep learning OCR pipeline for handwritten Sanskrit/Devanagari manuscripts, designing end-to-end recognition components and achieving 97% character-level accuracy with a 13-layer CNN in TensorFlow/Keras
- Implemented word detection (Detectron2), letter segmentation, and character recognition modules with custom preprocessing and augmentation; annotated COCO-format data in Label Studio for model training and evaluation

Indian Institute of Science Education and Research

Pune, India

Data Science Intern

March 2023 – November 2023

- Processed 400 GB of multi-source satellite imagery (INSAT-3D, MODIS Aqua/Terra) into ML-ready Land Surface Temperature datasets with under 5% cloud cover, designing subsetting, regridding, and collocation pipelines in Python (Xarray, SciPy)
- Built linear and nearest-neighbor interpolation pipelines (SciPy griddata) to align heterogeneous NetCDF/HDF grids, and implemented quality-control and validation logic with visualization to verify output accuracy

PROJECTS

Hate Speech Unlearning for LLMs | *PyTorch, LLaMA-2/3, LoRA, Hugging Face, Unsloth*

- Reimplemented the Selective Knowledge Unlearning (SKU) framework on LLaMA-2/3, integrating a contrastive learning module (InfoNCE loss) for harmful-prompt disentanglement and fine-tuning with cross-entropy and negative KL divergence losses on PKU-SafeRLHF/TruthfulQA, achieving a 46% drop in harmful response rate (62.6% → 33.9%)
- Designed a novel relearning stage using task-vector extraction and negation to restore control-task performance post-unlearning without reintroducing harmful behaviors; optimized training under hardware constraints via quantization and LoRA on 7B and 8B variants

Fast-Hack: GGUF K-Quant Content-Injection Attacks | *PyTorch, Hugging Face, LoRA (PEFT), llama.cpp, GGUF*

- Developed a six-step pipeline for content-injection attacks exploiting the gap between full-precision and GGUF k-quant LLM behavior in llama.cpp/Ollama, chaining LoRA supervised fine-tuning with closed-form weight interpolation to replace the brittle PGD approach of prior work
- Achieved up to 83% quantized attack-success rate at near-zero full-precision detection across two instruction-tuned models and five k-quant targets, running end-to-end on a single consumer GPU in under two hours per target

TECHNICAL SKILLS

Programming / Scripting: Python, C/C++, SQL (Proficient); Java (Familiar)

ML / AI: LLM Fine-tuning, RLHF/RFT, Machine Unlearning, Alignment & Safety, Adversarial Robustness, Model Evaluation, LoRA/QLoRA, Quantization, Chain-of-Thought Reasoning, Agentic AI (MCP)

Frameworks and Tools: PyTorch, Hugging Face, Unsloth, Detectron2, OpenCV, NumPy, Pandas, Matplotlib, LangChain, AWS, GCP, Git, Linux, Jupyter, HPC (USC CARC)

Soft Skills: Verbal and written communication, Problem-solving, Critical Thinking

PUBLICATIONS

Bhusari, A. and Kulkarni, A.J., 2025. Drone Path-Planning Leveraging Cohort Intelligence Algorithm. In *Optimization Techniques for Sustainable Environment Under Uncertainty* (pp. 135–147). Singapore: Springer Nature Singapore.